

Water Rocket Simulator

Standard

Advanced

Dark Shadow

Charlie

Reset to Default

Rocket

Rocket Outside Diameter (mm)

85

Nozzle Exit Diameter (mm)

7

Total Volume (L)

1.5

Water Amount (L)

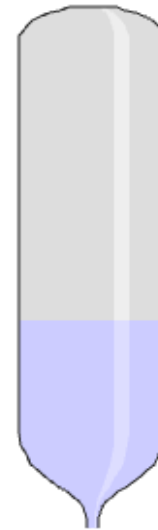
0.5

Empty Mass (g)

198

Drag Coefficient

0.50



Launcher

Pressure (bar)

8

116 psi

Launch Tube Diameter (mm)

0

Launch Tube Length (mm)

0

Run Simulation

Simulation Results

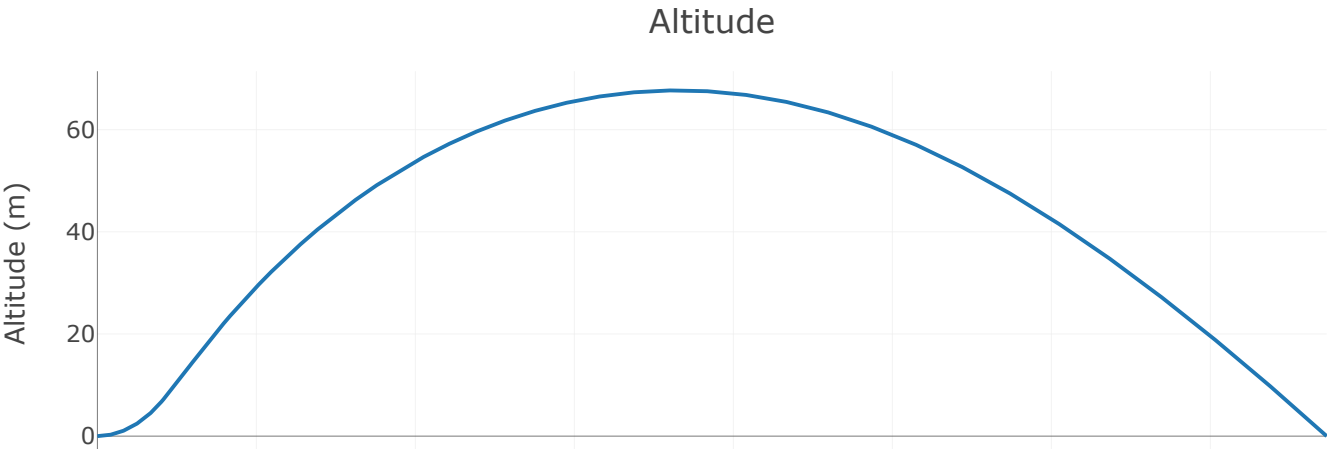
Burnout	
Time	0.72 s
Altitude	19.28 m
Velocity	38.37 m/s
Acceleration	-2.27 G

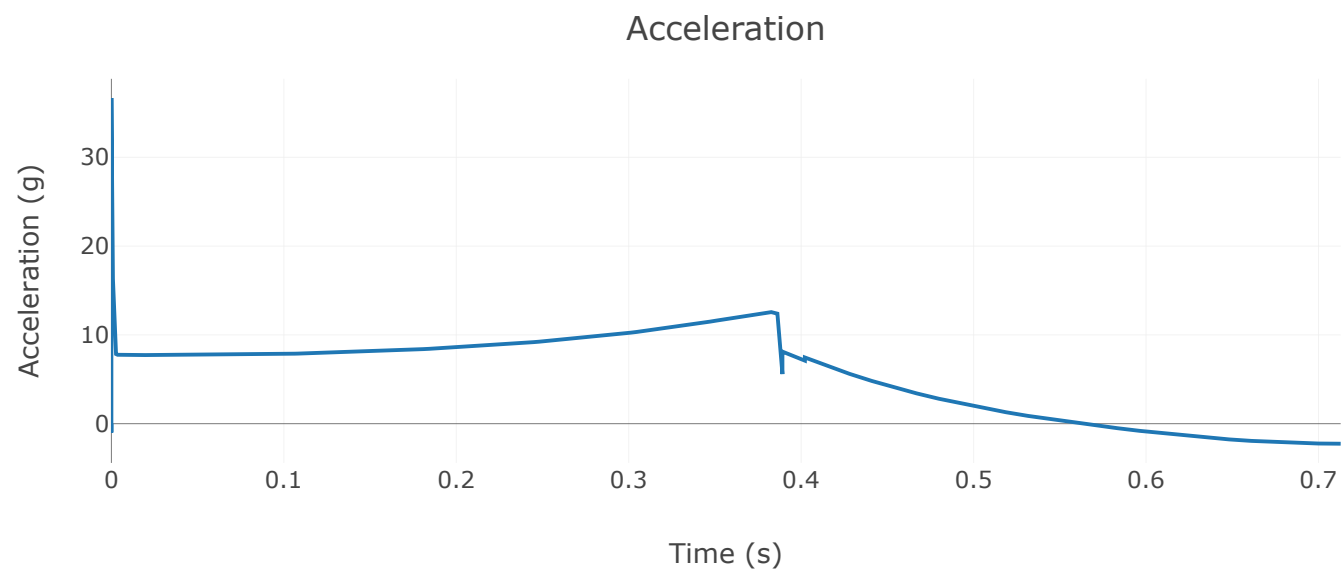
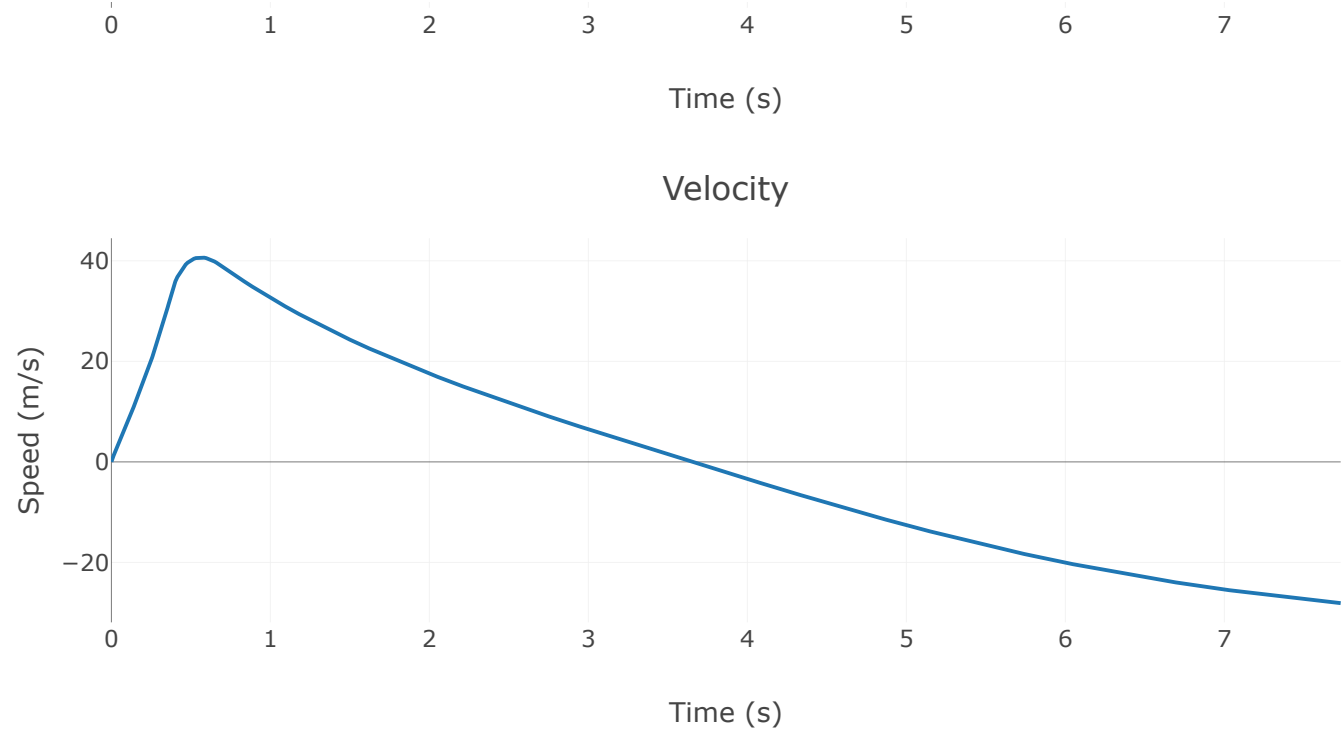
Apogee	
Time	3.65 s
Altitude	67.70 m (222 feet)

Thrust	
Total Impulse	19.04 Ns
Burn Time	0.60 s
Peak Thrust	60.48 N
Average Thrust	31 N
Motor Equivalent	D31

Crashdown	
Time	7.73 s
Velocity	-28.10 m/s

Graphs





Thrust